

SUMMARY

Aerospace Engineer with a background in satellite testing, systems engineering, and railgun development. Experienced in payload integration, hardware-in-the-loop testing, Python programming, and modeling and simulation. Eager to apply my technical expertise to impactful space programs and collaborate with innovative teams. Willing to travel and relocate.

EXPERIENCE

Rocket Lab (SolAero) – Albuquerque, NM**2024-present***Senior R&D Engineer – Spacecraft Solar Cell Qualification and Reliability*

- Led qualification efforts for emerging solar cell technologies, ensuring compliance with industry standards and customer requirements.
- Planned and executed testing campaigns, including electrical characterization, irradiation, thermal shock, and humidity exposure. Consulted with subject matter experts to ensure “test like you fly” principles were followed.
- Developed schedules and coordinated resources to meet program milestones. Collaborated with external labs to reduce costs and accelerate timelines while maintaining quality.
- Produced detailed qualification reports for internal and external stakeholders to support solar panel development, design, and integration.
- Established standardized testing guidelines for electrostatic discharge and atomic oxygen effects, enabling consistent evaluation processes and aiding product development.

Amazon (Project Kuiper) – Redmond, WA**2021-2023***Systems and Test Engineer – Satellite Avionics*

- Defined and managed requirements for unit acceptance testing, ensuring alignment with design specifications and flow down into test automation software.
- Led cost reduction initiatives for test system equipment by optimizing for price, commonality, design complexity, labor, and lead times. Achieved significant cost savings while maintaining technical performance.
- Developed and maintained schedules to track test system bring-up and early production testing. Coordinated with customers to resolve dependencies and track milestones.

Hardware Automation and Test Engineer – Propulsion System

- Led environmental and integration testing for the electric thruster unit. Designed, built, and operated test racks to validate flight hardware functionality in a range of simulated spacecraft environmental conditions.
- Designed hardware-in-the-loop (HIL) emulators to enable rapid development of the thruster system. Emulated endpoints included the anode, cathode, torque rods, valves, and heaters.
- Built an automated test framework in Python to handle data recording, pass/fail analysis, and report generation. The code could run multi-day tests without errors or human intervention.
- Developed Python equipment drivers to interface with test rack hardware, including power supplies, electronic loads, relays, thermal chambers, DMMs, and SMUs.

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Sandia National Laboratories – Albuquerque, NM

2020-2021

Guidance, Navigation, and Control Engineer – Hypersonic Missile

- Implemented novel guidance laws in the CUDA C++ hypersonic missile simulator, utilizing Monte Carlo analysis to evaluate GNC performance.
- Developed a digital twin for the hypersonic missile avionics, modeling components and sensors in Simulink. Modularized components to enable rapid hardware-in-the-loop testing and provide a debugging platform for avionics software.
- Led testing of fin and actuator sub-assemblies, including writing test plans, operating equipment, analyzing results, and implementing safety procedures.

General Atomics

2018-2020

Railgun Development Engineer

- Collected and analyzed in-bore railgun projectile telemetry data to improve railgun physics models, including a mechanism for contact metal phase transition.
- Created a railgun bore scanner, leveraging a laser sheet and camera to generate 3D cross-sectional representations of the bore, highlighting wear over time and cleaning effectiveness.
- Designed a projectile tracking system using stereo vision and high-speed IR cameras, achieving radar-like accuracy. Repurposed this system for shrapnel tracking during dispense testing, eliminating manual witness card analysis.
- Automated the identification and tracking of imperfections during wound-barrel composite manufacturing, improving quality control and enabling continuous manufacturing.
- Conducted radar performance tuning using UAVs equipped with RTK GPS. Enabled radar to transmit real-time location data to the railgun projectile.

Aerospace Engineer – Proposal Development

- Formulated advanced control methods (PID, PTOC, and SMC) for the Next Generation Interceptor, comparing performance using a MATLAB-based simulator.
- Developed a thermal fluid simulator for a thermal management system in a high-powered laser, analyzing different layouts, fluids, and heat exchangers to maximize runtime.
- Evaluated underwater buoyancy glider submarine designs using CFD, utilizing scale model testing to validate the simulations. Results were used to define the set of mission profiles.

University of Illinois

2015-2018

Researcher and Teaching Assistant

- Teaching assistant for the spacecraft electric propulsion class. Topics covered included plasma physics, Hall, gridded ion, pulsed plasma, MPD, arcjet, and resistojet thrusters.
- Research assistant in the Helicon Injected Inertial Plasma Electrostatic Rocket (HIIPER) electric propulsion lab. Operated the thruster, rebuilt the helicon, and designed a micronewton thrust stand.
- Research assistant in the Center for Plasma-Material Interactions Fusion lab. Worked on the tokamak and theta-pinch devices and tin-lithium material study.

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EDUCATION

University of Illinois at Urbana-Champaign **GPA: 4.00** **2018**

Master of Science, Aerospace Engineering

Focused on: electric propulsion, combustion, distributed and satellite control systems

University of Illinois at Urbana-Champaign **GPA: 3.97** **2017**

Bachelor of Science, Aerospace Engineering

Focused on: control systems, CFD, systems engineering, UAVs, thermodynamics

KEY SKILLS

- Systems Engineering, Program Management, Payload Integration
- Spacecraft Systems Qualification and Testing
- Hardware-in-the-Loop (HIL) Testing
- Technical Documentation and Reporting
- Python, C++, MATLAB, Simulink, SolidWorks
- Linux, Windows, Git, Jira, Confluence